

1. Understanding the Business Problem

The main purpose of this project is to understand customer shopping behaviour and identify the factors that influence their purchasing decisions. The retail company wants to analyse how customers interact with different products, categories, and shopping channels, including online and offline platforms.

2. Identifying Customer Purchase Patterns

The company has noticed changes in customer preferences and buying habits. Therefore, this analysis focuses on finding patterns related to factors such as discounts, customer reviews, seasonal trends, and payment methods that may affect customer choices and repeat purchases.

3. Purpose of Data Analysis

By analysing consumer behaviour data, the project aims to generate meaningful insights that can help the company improve customer satisfaction, strengthen customer relationships, and make better decisions regarding marketing and product strategies.

4. Data Preparation and Modelling Using Python

The first stage of the project involves preparing the raw dataset for analysis. This includes cleaning the data, handling missing values, transforming variables, and creating a suitable structure that can be used for further analysis.

5. Data Analysis Using SQL

After preparing the data, SQL will be used to organise information into a structured format and analyse customer segments, purchasing behaviour, loyalty patterns, and the factors influencing buying decisions.

6. Data Visualisation and Insights Using Power BI

An interactive Power BI dashboard will be developed to present important trends and patterns in a clear visual format. The dashboard will help stakeholders easily understand customer behaviour and support data-driven decision-making.

7. Business Report and Recommendations

The final findings will be summarised in a project report and presentation. The report will explain key insights from the analysis and provide practical recommendations that can help the company improve marketing approaches, customer engagement, and product planning.

8. Project Documentation and GitHub Repository

All project materials, including Python scripts, SQL queries, and Power BI dashboard files, will be organised in a structured GitHub repository. This will ensure proper documentation and make the project easier to review and maintain.